Questions

Q1.

A student investigates moments of forces.

Figure 14 shows the apparatus used.

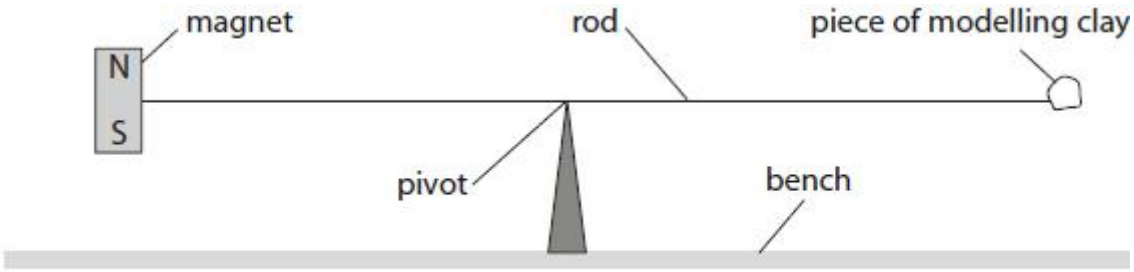

Figure 14

The pivot is under the centre of the rod.

A magnet is fixed to one end of the rod.

A piece of modelling clay is fixed to the other end of the rod.

The system is in equilibrium.

(a) The student fixes a coil to the bench under the magnet as shown in Figure 15.

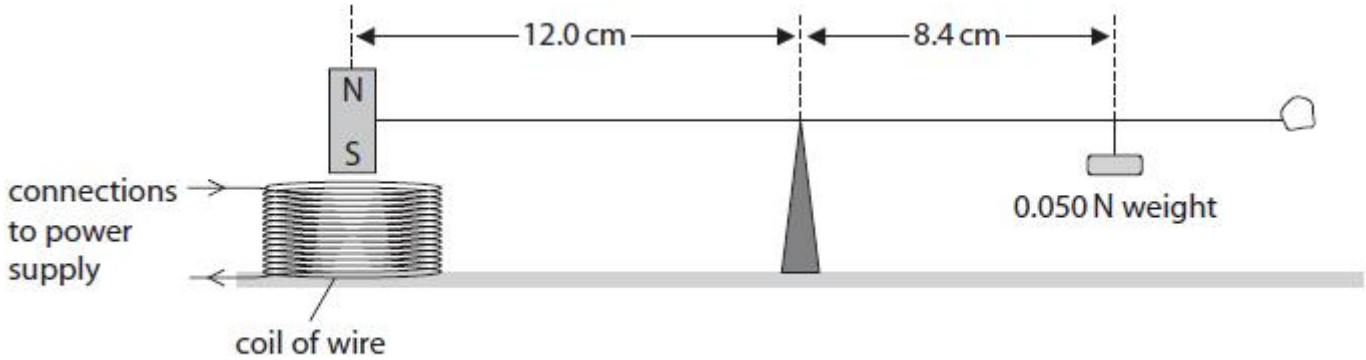

Figure 15

The coil of wire is connected to a d.c. power supply so that there is a current in the coil.
To bring the system back into equilibrium, the student hangs a 0.050 N weight on the rod, 8.4 cm away from the pivot, as shown in Figure 15.
Calculate the size of the force between the magnet and the coil.

(3)

force = ... N

(b) Describe how the student could develop the investigation to determine if the size of the force between the magnet and the coil is directly proportional to the size of the current in the coil.

(4)

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(Total for question = 7 marks)

Q2.

Figure 14 show a rack and pinion system, used on a mountain railway.

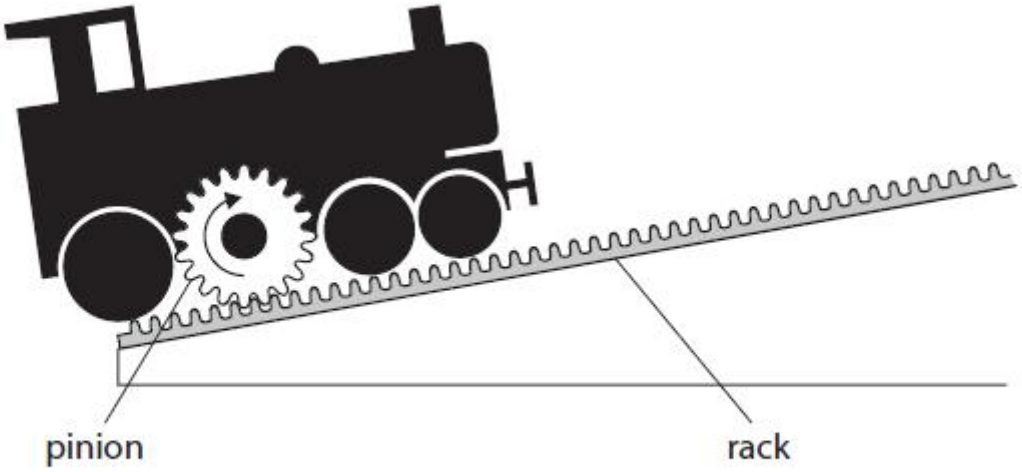

Figure 14

The teeth on the rack are 8.0 cm apart.

Calculate how far along the rack the train moves when the pinion turns through one complete revolution.

(2)

distance = ... m

(Total for question = 2 marks)

Q3.

The extension of the spring in Figure 16b is now determined for different weights added to the spring up to a maximum of 6 N.

Figure 17 shows a graph of the results.

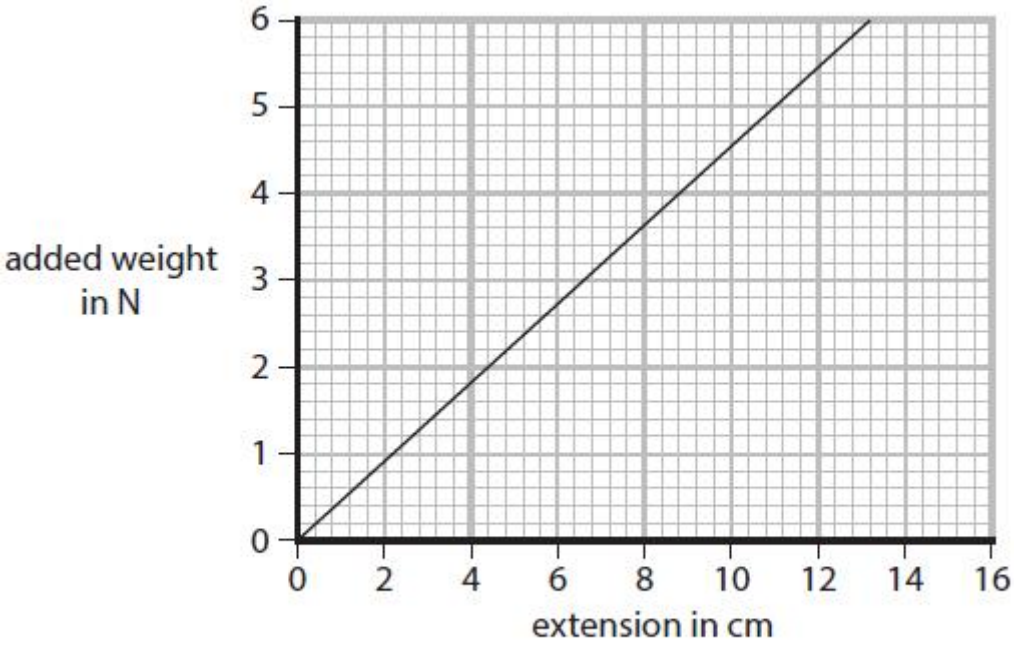

Figure 17

(i) The increase in potential energy stored in the spring is equal to the area under the graph in Figure 17.

Calculate the increase in potential energy stored in the spring when the spring is extended from 0 cm to 8 cm.
Give your answer in joules (J).

(3)

energy = ... J

(ii) Calculate the spring constant of the spring using data from the graph in Figure 17.

State the unit.

(2)

spring constant = ... unit .....

(iii) The box in Figure 18 represents a weight added to the end of the spring.

Figure 18

Draw **two** vectors on Figure 18 to complete a free body force diagram of a 5 N weight in equilibrium.

(2)

(Total for question = 7 marks)

Q4.

Figure 3 is a diagram of the forces acting on a swing.

The swing is not moving.

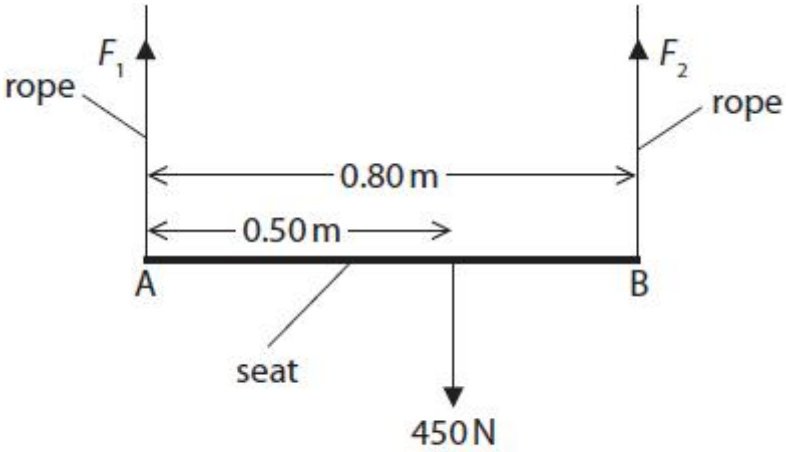

Figure 3

The seat of the swing, AB, is 0.80 m long.

A person of weight 450 N sits on the seat.

The person's weight acts at a distance of 0.50 m from A as shown in Figure 3.

Ignore the weight of the seat.

The upward forces exerted by the ropes on the seat are F_1 and F_2.

Calculate the force F_2 by taking moments about A.

(3)

force F_2 = ... N

(Total for question = 3 marks)

Q5.

Figure 8 shows a bicycle.

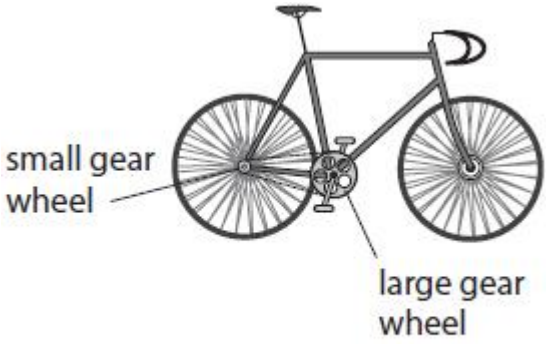

Figure 8

(i) The rider uses the pedals to make the large gear wheel turn.

The large gear wheel moves the chain.
The chain turns the small gear wheel.
The large gear wheel has 48 teeth.
The small gear wheel has 12 teeth.
The large gear wheel turns 2 times each second.
Calculate the number of times that the small gear wheel turns each second.

(2)

..... turns each second

(ii) Oil is applied to the wheel of a bicycle at the point shown in Figure 9.

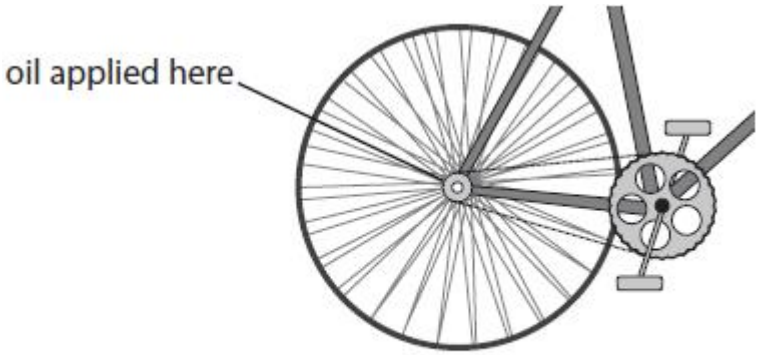

Figure 9

Explain how the oil improves the efficiency of the bicycle.

(3)

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(Total for question = 5 marks)

Q6.

Figure 13 shows a toy that a student makes for a nursery school.

for a nursery school. The rod hangs by a string from the ceiling.

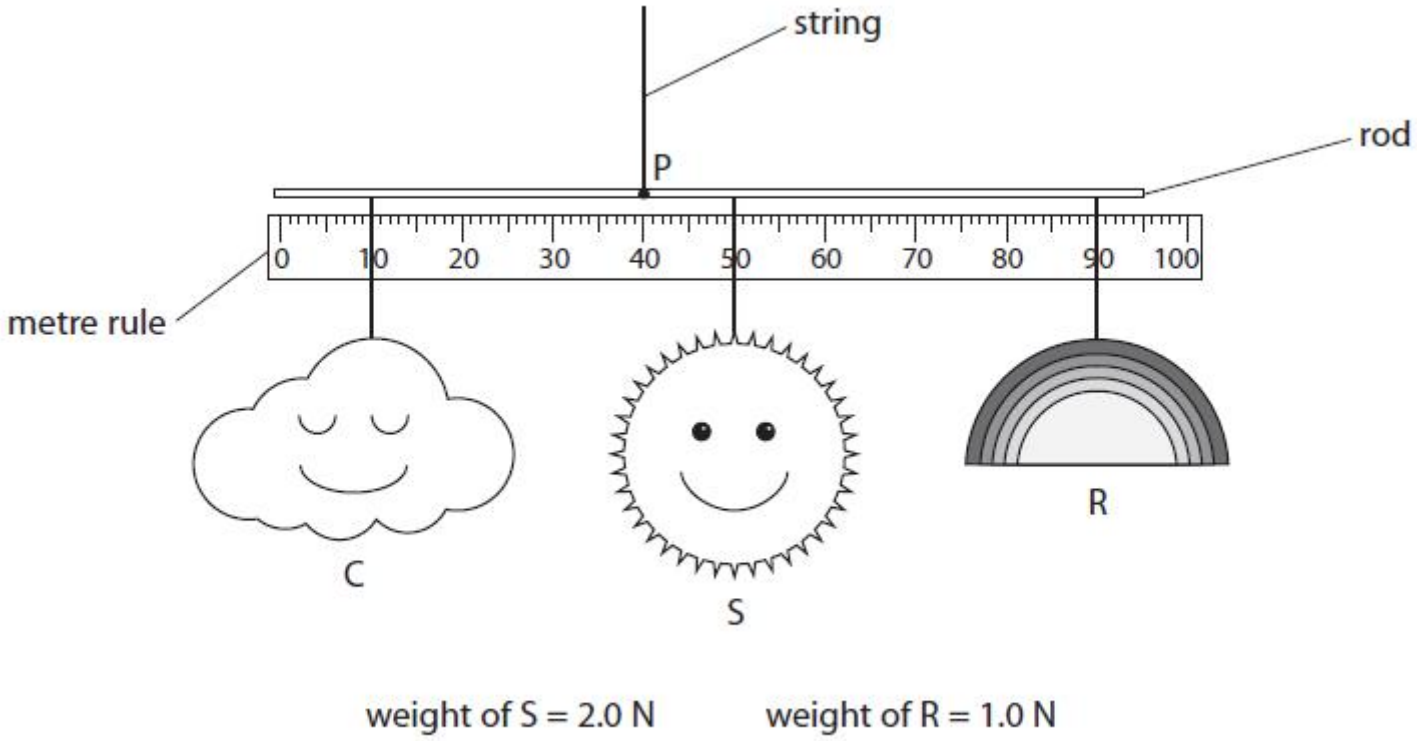

Figure 13

(i) The moment of C about point P is 0.60 Nm.

Calculate the weight of C.

Use the equation

$$\text{moment} = F \times d$$

(2)

weight of C ..... N

(ii) Show that the total moment of S and R about P is 0.70 N m.

(2)

(iii) Using the data in the question and the principle of moments, determine if the toy shown in Figure 13 is in equilibrium.

The rod is very light so its weight can be ignored.

(3)

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(Total for question = 7 marks)

Q7.

Figure 12 shows three toy animals hanging from a rod.

The rod hangs from the ceiling by a string tied to the centre of the rod.

The system is in equilibrium.

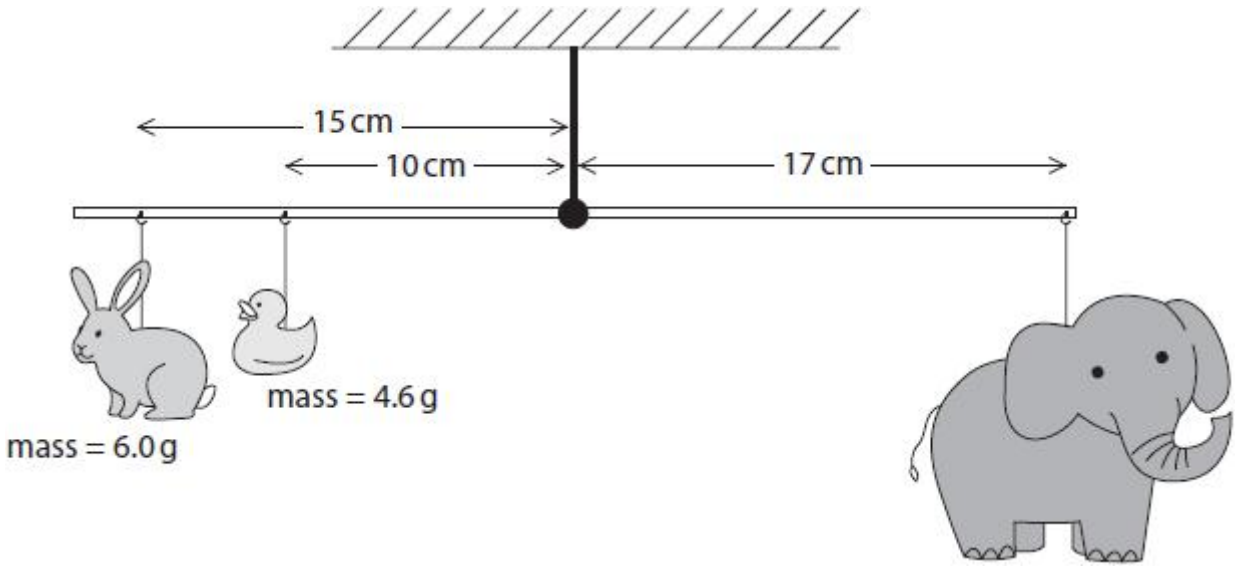

Figure 12

Use the principle of moments to calculate the mass of the toy elephant.

(4)

mass = ... g

(Total for question = 4 marks)

Q8.

Figure 19 shows two forces, P and Q, acting at point X.

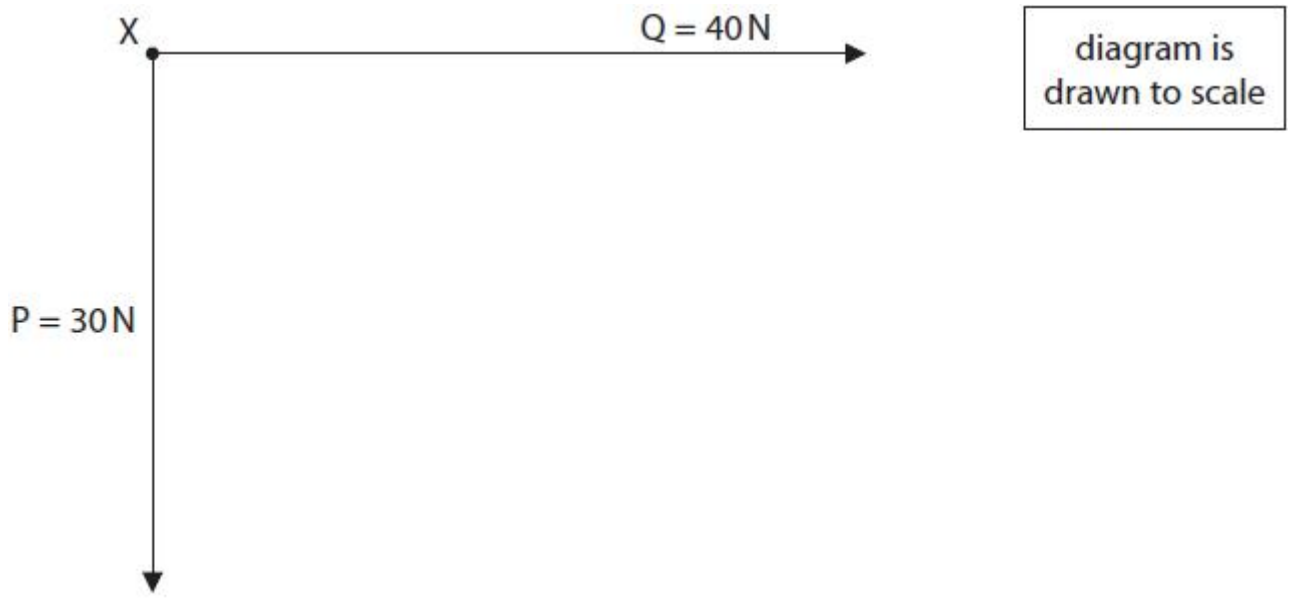

Figure 19

Complete the diagram in Figure 19 to show the size and direction of the resultant force, R, on point X.

(Total for question = 2 marks)

Q9.

Figure 19 shows two forces, P and Q, acting at point X.

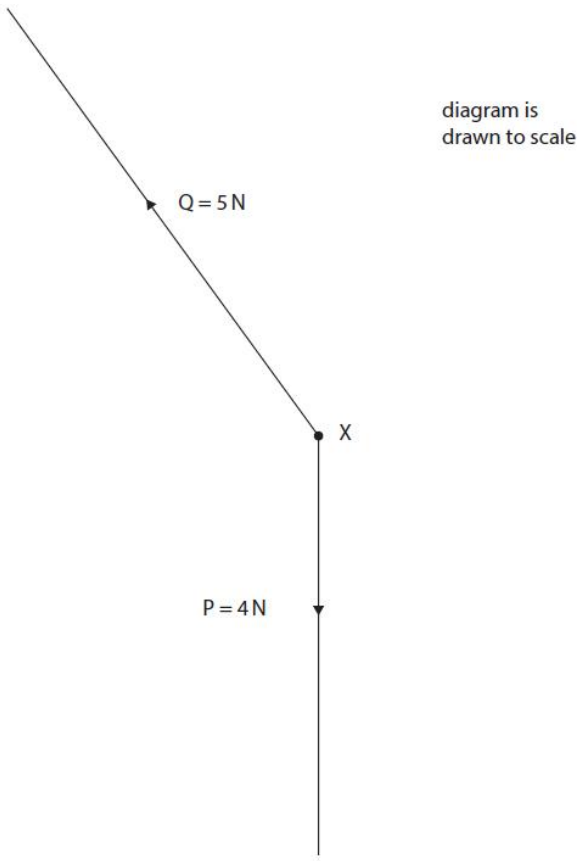

Figure 19

Complete the diagram in Figure 19 to show the size and direction of the resultant force, R, on point X.

(2)

(Total for question = 2 marks)

Q10.

Figure 14 shows a demolition ball of mass 400 kg.

The ball is used to demolish a wall.

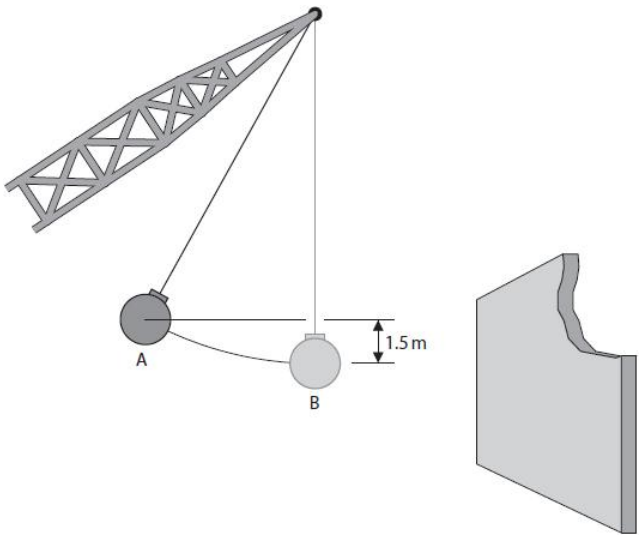

Figure 14

As the ball swings from A to B, it moves through a vertical height of 1.5 m.

(i) Calculate the change in gravitational potential energy.

Take gravitational field strength, g, to be 10 N/kg.

(3)

(ii) The heavy ball comes to rest soon after smashing into the wall. In doing this, the temperatures of the ball, wall and surroundings all increase slightly.

Explain this observation.

(2)

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(iii) Which of these quantities of the ball changes in **both** magnitude and direction while the ball is swinging?

Put a cross (☒)in the box next to your answer.

(1)

- ☐ **A** gravitational potential energy
- ☐ **B** velocity
- ☐ **C** the gravitational force acting
- ☐ **D** kinetic energy

(Total for question = 6 marks)

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
a	recall and substitution (1) (force x 12.0 =) 0.050 x 8.4 rearrangement (1) (force =) $\frac{0.050 \times 8.4}{12.0}$ evaluation (1) (force =) 0.035 (N)	allow substitution and rearrangement in either order award full marks for the correct answer without working. if no other marks scored then award 1 mark for answers that round to 29 (eg 28.57) (substitution mark)	**(3)** **AO2**
Question number	**Answer**	**Additional guidance**	**Mark**
b	a description to include **four** of the following measure the value of current (1) measure force or distance(1) vary the current (1) restore equilibrium of system (1) calculate ratio between force and current or distance and current (1) if ratio is the same then they are proportional (1)	accept calculate for measure increase weight or move (existing) weight to new position plot a graph of force / distance against current graph would be a straight line (through the origin)	**(4)** **AO3**

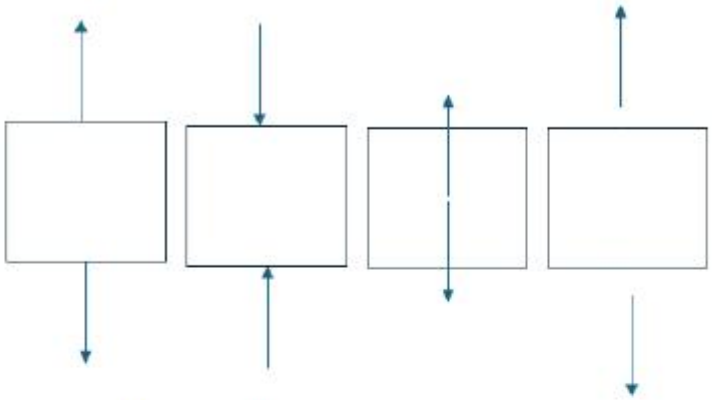
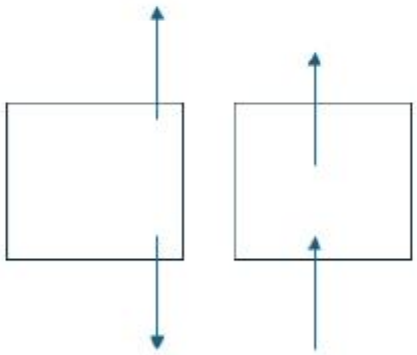
Q2.

Question number	Answer	Additional guidance	Mark
	counting teeth on the pinion (1) evaluation (1) 1.6 (m)	allow between 18 and 22 inclusive 20 x 0.08 ecf number of teeth answer in range 1.44 to 1.76 scores 2 marks award full marks for the correct answer without working power of 10 error scores 1 mark maximum	**(2)** **AO3.1**

Q3.

Question number	Answer	Additional guidance	Mark
(i)	relevant values from graph(1) 8 ($\times 10^{-2}$) and 3.6 seen substitution (1) (energy =) 0.5 x 3.6 x 8 ($\times 10^{-2}$) evaluation (1) (energy =) 0.14 (J)	 accept 8 ($\times 10^{-2}$) and values between 3.6 and 3.7 accept value between 0.14 and 0.15 (J) values between 0.14 and 0.15 (J) to any other power of ten scores 2 marks values between 0.28 and 0.30 (J) scores 2 marks values between 0.28 and 0.30 (J) to any other power of ten scores 1 mark award full marks for correct answer without working	**(3)** **AO2**

Question number	Answer	Additional guidance	Mark
(ii)	45 or 0.45 (1) **OR** 45 N/m or 0.45 N/cm (2)	allow values between 44 and 47 or between 0.44 and 0.47 allow full marks for values between 44 and 47 N/m or between 0.44 and 0.47 N/cm	**(2)** **AO2**

Question number	Answer	Additional guidance	Mark
(iii)	two straight vertical arrows in opposite directions and (if extended) go through centre of the box (judge by eye) (1) label 5N on both arrows (1)	examples  scores this mark  does not score this mark independent mark	**(2)** **AO1**

Q4.

Question number	Answer	Additional guidance	Mark
	any correct moment (1) 450 x 0.50 or 225 or 0.80 x F_2 substitution into prin. of moment equation (1) 450 x 0.50 = 0.80 x F_2 evaluation (1) 280 (N) (for question at end)	 allow 450 x 0.3 moment taken about B allow statement of prin. of moments accept numbers which round to 280 such as 281.25 award full marks for correct answer without working.	(3)

Q5.

Question Number	Answer	Additional guidance	Mark
(i)	(In every second), distance moved by chain around large gear = distance moved by chain around small gear (1) 2 x 48 = turns x 12 rearrangement and evaluation (1) 8 (turns each second)	accept use of gear ratio seen or implied e.g. 4:1 or 4/1 or 48:12 or 48/12 or converse e.g. 1:4 award full marks for the correct answer without working	(2)

Question Number	Answer	Additional guidance	Mark
(ii)	An explanation linking reduces friction/amount of thermal energy transferred (1) extra useful energy is available/less input energy is required (1) efficiency = useful energy transferred (by the bicycle) ÷ total energy supplied (to the bicycle) (1)	 (oil provides) lubrication less energy wasted allow for the last **two** mark points; **either** less input energy is required to produce the same output for **2** marks **or** more output energy is available for the same input energy for **2** marks	**(3)**

Q6.

Question number	Answer	Additional guidance	Mark
(i)	substitution and rearrangement (1) (force = $\frac{\text{moment}}{\text{distance}}$) = $\frac{0.6}{3(\times 10^{-1})}$ evaluation (1) 2(.0) (N)	 reject 0.6 x 3 = 1.8 award full marks for the correct answer without working ignore significant figures 2(.0) to any other power of ten scores 1 mark maximum	**(2)** **AO2.1**

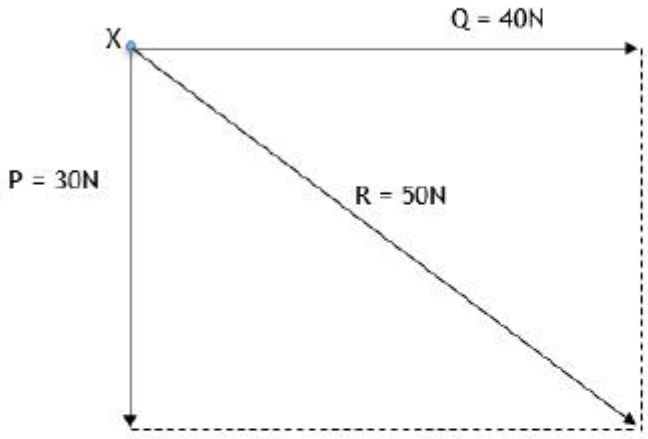
Question number	Answer	Additional guidance	Mark
(ii)	 correct calculation of one moment (1) correct calculation of second moment and adding of moments seen (1)	**show that** question either 2x0.1 or 1x0.5 seen 2(.0)x0.1(0) + 1(.0)x0.5 scores 2 marks 0.2+0.5 scores 2 marks accept calculations in alternative units (e.g. N cm) if correct conversion(s) seen if no other marks scored, the addition of two other moments can score 1 mark maximum	**(2)** **AO2.1**

Question number	Answer	Additional guidance	Mark
(iii)	explanation linking **three** from: **{sum** of / **total }** clockwise moments = **{sum** of / **total }** anticlockwise moments (1) for a system in equilibrium / balance (1) clockwise and anticlockwise moments compared (1) **so** rod not in equilibrium (1)	about the same point / about a point e.g. clockwise moment > anticlockwise moment or reverse argument 0.7 > 0.6 0.7 ≠ 0.6 rod will **rotate clockwise** MP4 can only be scored if MP3 awarded	**(3)** **AO3.2**

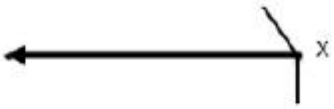
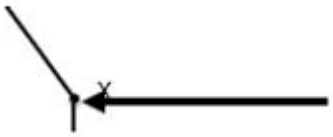
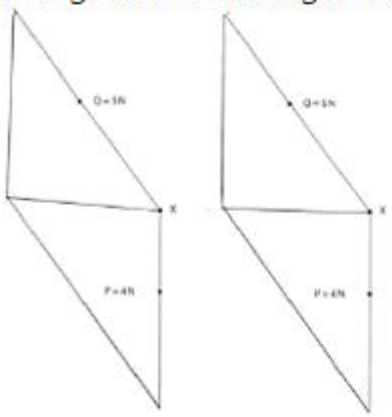
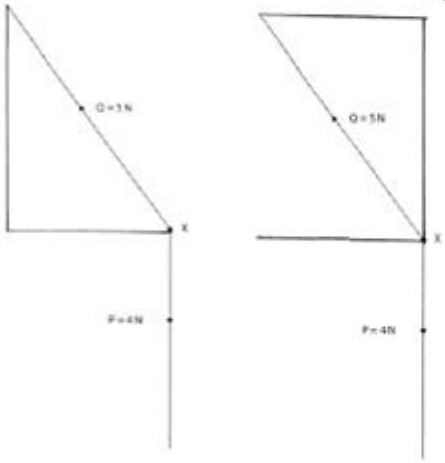
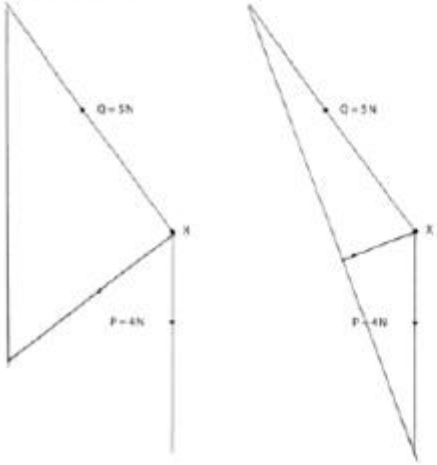
Q7.

Question Number:	Answer	Additional Guidance	Mark
	recall clockwise moment = anticlockwise moment (1) moment = force x (perpendicular) distance (1) substitution (1) $m \times 17 = (6 \times 15) + (4.6 \times 10)$ rearrangement and evaluation (1) m = 8.0 (g)	calculations need not include *g* (which cancels out from all terms) substitution and rearrangement in either order $m \times 17 = 90+46$ $m = \frac{(6\times15)+(4.6\times10)}{17}$ m = 136/ 17 award full marks for correct answer without working	**(4)** AO 1 1 AO 2 1

Q8.

Question number	Answer	Additional guidance	Mark
		independent marks	**(2)** **A01**
	<u>arrow</u> (any length) (labelled R) in correct direction (judge by eye) (1)	 construction lines need not be shown arrow head must be present for MP1	
	(size of R =) 50N (1)	accept answers in range 48N to 52N obtained from scale drawing working need not be shown	

Q9.

Question	Answer	Additional guidance	Mark
	correct direction (1) any size horizontal (judge by eye) arrow pointing left at point x  or  	if arrow missing or incorrect then allow attempt to find direction by construction of a suitable parallelogram or triangle e.g.  or  but not  	**(2)** **AO2**

	indication of size (1) drawn correctly, size judged by eye or any arrow labelled 3 (N) . Allow values between 2.5 and 3.5	independent mark or 	

Q10.

Question number	Answer	Additional guidance	Mark
(i)	Recall GPE = m x g x Δh (1) Substitution = 400 x 9.8 x 1.5 (1) Evaluation = 5900 (J) (1) (which is nearly 6000 J)	accept 5880 (J)	**(3)**

Question number	Answer	Additional guidance	Mark
(ii)	An explanation that combines identification – knowledge (1 mark) and reasoning (1 mark) • energy is dissipated/scattered (1) • into the surroundings (1)	energy from a loss of ball's PE / its gain in KE ends up as (kinetic) energy of molecules (of ball / wall / air)	**(2)**

Question number	Answer	Mark
(iii)	**B** velocity	**(1)**